

Bio334

Phylogenetic reconstruction Part I: Distance-based methods

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Setup for today

- Exercises: `Bio334/04_njtrees/exercises`
- If you haven't cloned the repository yet:
 - Open the terminal, ``cd`` to a directory of your choice, then type
``git clone``
`https://github.com/meringlab/Bio334``
- If you cloned it yesterday:
 - make sure to type ``git pull`` within the Bio334 folder to get the latest updates

Exercise 1 – Questions

- Complexity

$$\binom{n}{2} \sim O(n^2)$$

Time: 1000 proteins $\rightarrow$ 1 s

10.000 proteins $\rightarrow 10s * 10s = 100$ s

100.000 proteins $\rightarrow 100 * 100 = 10.000s$ ($\sim 3h$)

$10^9 \rightarrow 31.700$ years

Exercise 1 – Questions

- Shortcomings of naïve Jaccard

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- Shortcomings of naïve Jaccard
 - All amino acid transitions have equal probability:
unrealistic
 - Use explicit substitution matrices (BLOSUM, PAM)

PAM Substitution Matrix

	A	R	N	D	C	Q	E	G	H	I	L	K	M	F	P	S	T	W	Y	V
A	4	-3	-1	-1	-3	-2	0	1	-3	-2	-3	-3	-2	-5	1	1	1	-7	-4	0
R	-3	7	-2	-4	-5	1	-3	-5	1	-3	-5	2	-1	-6	-1	-1	-3	1	-6	-4
N	-1	-2	5	3	-5	-1	1	-1	2	-3	-4	1	-4	-5	-2	1	0	-5	-2	-3
D	-1	-4	3	5	-7	0	4	-1	-1	-4	-6	-1	-5	-8	-3	-1	-2	-9	-6	-4
C	-3	-5	-5	-7	9	-8	-8	-5	-4	-3	-8	-8	-7	-7	-4	-1	-4	-9	-1	-3
Q	-2	1	-1	0	-8	6	2	-3	3	-4	-2	0	-2	-7	-1	-2	-2	-7	-6	-3
E	0	-3	1	4	-8	2	5	-1	-1	-3	-5	-1	-4	-8	-2	-1	-2	-9	-5	-3
G	1	-5	-1	-1	-5	-3	-1	5	-4	-5	-6	-3	-4	-6	-2	0	-2	-9	-7	-3
H	-3	1	2	-1	-4	3	-1	-4	7	-4	-3	-2	-4	-3	-1	-2	-3	-4	-1	-3
I	-2	-3	-3	-4	-3	-4	-3	-5	-4	6	1	-3	1	0	-4	-3	0	-7	-3	3
L	-3	-5	-4	-6	-8	-2	-5	-6	-3	1	6	-4	3	0	-4	-4	-3	-3	-3	0
K	-3	2	1	-1	-8	0	-1	-3	-2	-3	-4	5	0	-7	-3	-1	-1	-6	-6	-4
M	-2	-1	-4	-5	-7	-2	-4	-4	-4	1	3	0	9	-1	-4	-3	-1	-6	-5	1
F	-5	-6	-5	-8	-7	-7	-8	-6	-3	0	0	-7	-1	8	-6	-4	-5	-1	4	-3
P	1	-1	-2	-3	-4	-1	-2	-2	-1	-4	-4	-3	-4	-6	7	0	-1	-7	-7	-3
S	1	-1	1	-1	-1	-2	-1	0	-2	-3	-4	-1	-3	-4	0	4	2	-3	-4	-2
T	1	-3	0	-2	-4	-2	-2	-2	-3	0	-3	-1	-1	-5	-1	2	5	-7	-4	0
W	-7	1	-5	-9	-9	-7	-9	-9	-4	-7	-3	-6	-6	-1	-7	-3	-7	12	-2	-9
Y	-4	-6	-2	-6	-1	-6	-5	-7	-1	-3	-3	-6	-5	4	-7	-4	-4	-2	9	-4
V	0	-4	-3	-4	-3	-3	-3	-3	-3	3	0	-4	1	-3	-3	-2	0	-9	-4	5



Exercise 1 – Questions

- Shortcomings of naïve Jaccard
 - All amino acid transitions have equal probability: unrealistic
 - Use explicit substitution matrices (BLOSUM, PAM)
 - Dependencies between neighboring sites (e.g. indels) are not considered
 - Smaller gap penalties, explicit modeling

Exercise 1 – Questions

- Nucleotides vs Amino acids

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 - Nucleotides
 - High resolution: small differences can be detected
 - More neutrally evolving regions: less biased similarity estimates
 - Distinction between synonymous and non-synonymous mutations

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- Nucleotides vs Amino acids
 - Nucleotides
 - High resolution: small differences can be detected
 - More neutrally evolving regions: less biased similarity estimates
 - Distinction between synonymous and non-synonymous mutations
 - Amino acids
 - Stable over longer evolutionary time frames, used to address questions on non-recent evolution

Exercise 1 – Questions

- Distance calculation:

Protein_Q:GDRVG RKFII WFSIL GTAPF ALWLP YAD-A DTTAI LVILI GFIIS SAFAS
Protein_A:GDRVG RKFII WFSIL G**A**APF ALWLP YAD-A **Q**TTAI L**I**VLI GFIIS SAFAS

1. Calculate Jaccard index J:

$$J(A,Q) = \frac{|A \cap Q|}{|A \cup Q|} = \frac{46}{50}$$

2. Calculate Jaccard distance:

$$d_j(A, Q) = 1 - J(A, Q) = 1 - \frac{|A \cap Q|}{|A \cup Q|} = 1 - \frac{46}{50} = 0.08$$

Exercise 2 – Questions

D	Q	A	B	C	D
Q	0	0.08	0.36	0.36	0.38
A	0.08	0	0.32	0.34	0.36
B	0.36	0.32	0	0.32	0.26
C	0.36	0.34	0.32	0	0.22
D	0.38	0.36	0.26	0.22	0

Exercise 2 – Questions

	Q	A	B	C	D
Q	0	0.08	0.36	0.36	0.38
A	0.08	0	0.32	0.34	0.36
B	0.36	0.32	0	0.32	0.26
C	0.36	0.34	0.32	0	0.22
D	0.38	0.36	0.26	0.22	0

➤ Compute matrix D2:

	QA	B	C	D
QA	0			
B		0	0.32	0.26
C		0.32	0	0.22
D		0.26	0.22	0

Exercise 2 – Questions

	Q	A	B	C	D
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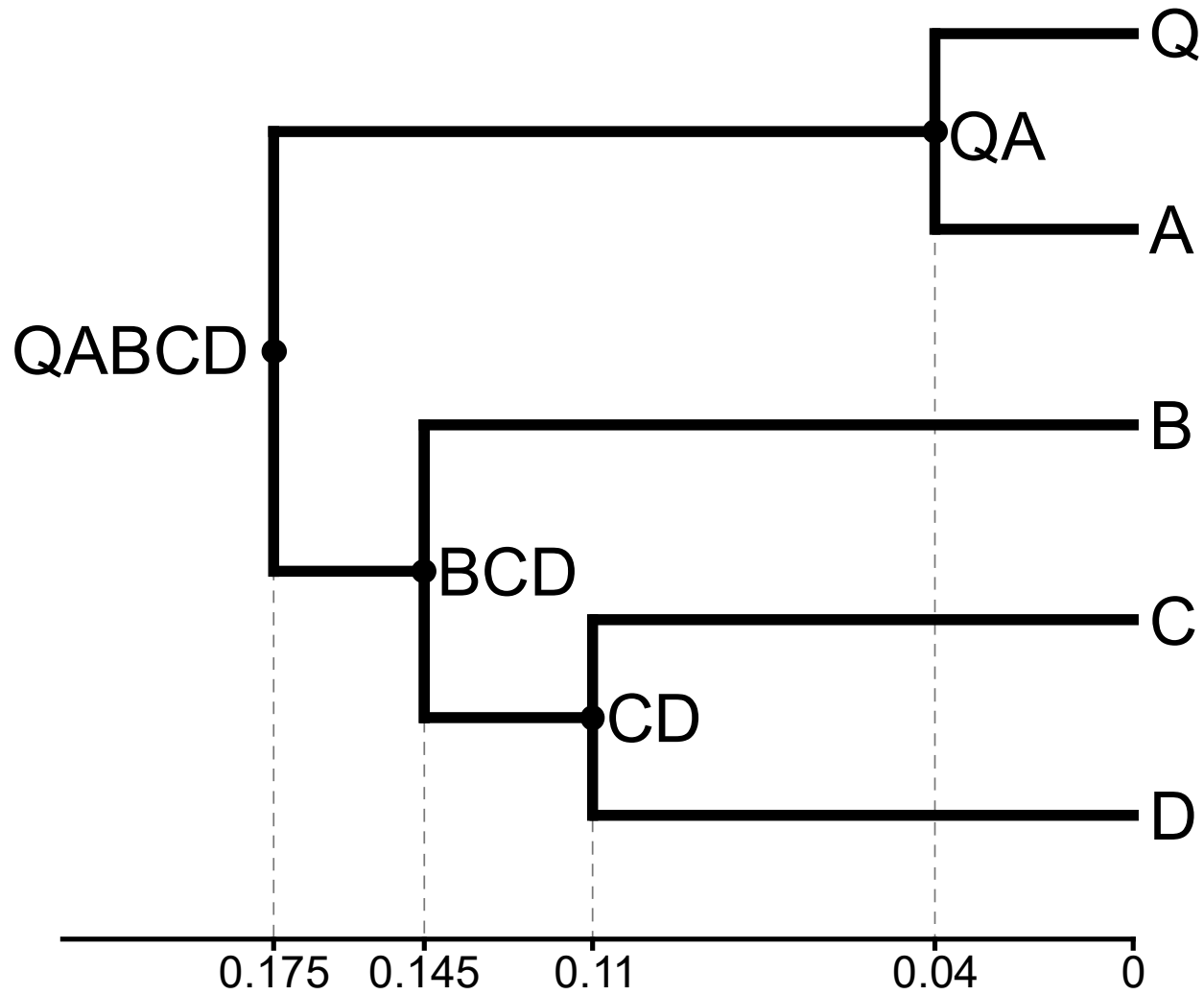
– for distance between **QA** and C:

➤ mean between all node distances of QA and C:

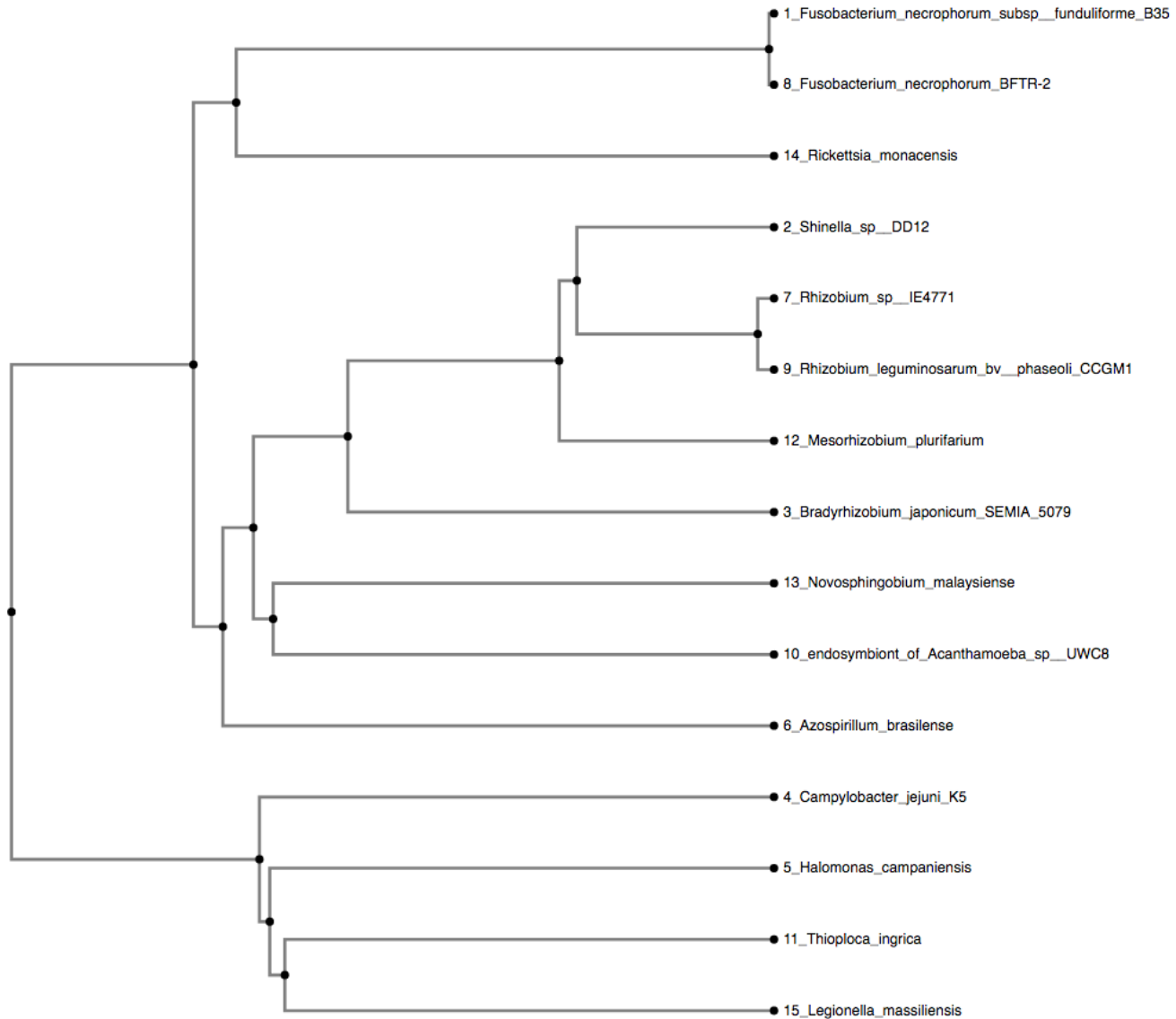
$$d_j(QA, C) = \frac{d_j(Q, C) + d_j(A, C)}{2} = \frac{0.36 + 0.34}{2} = 0.35$$

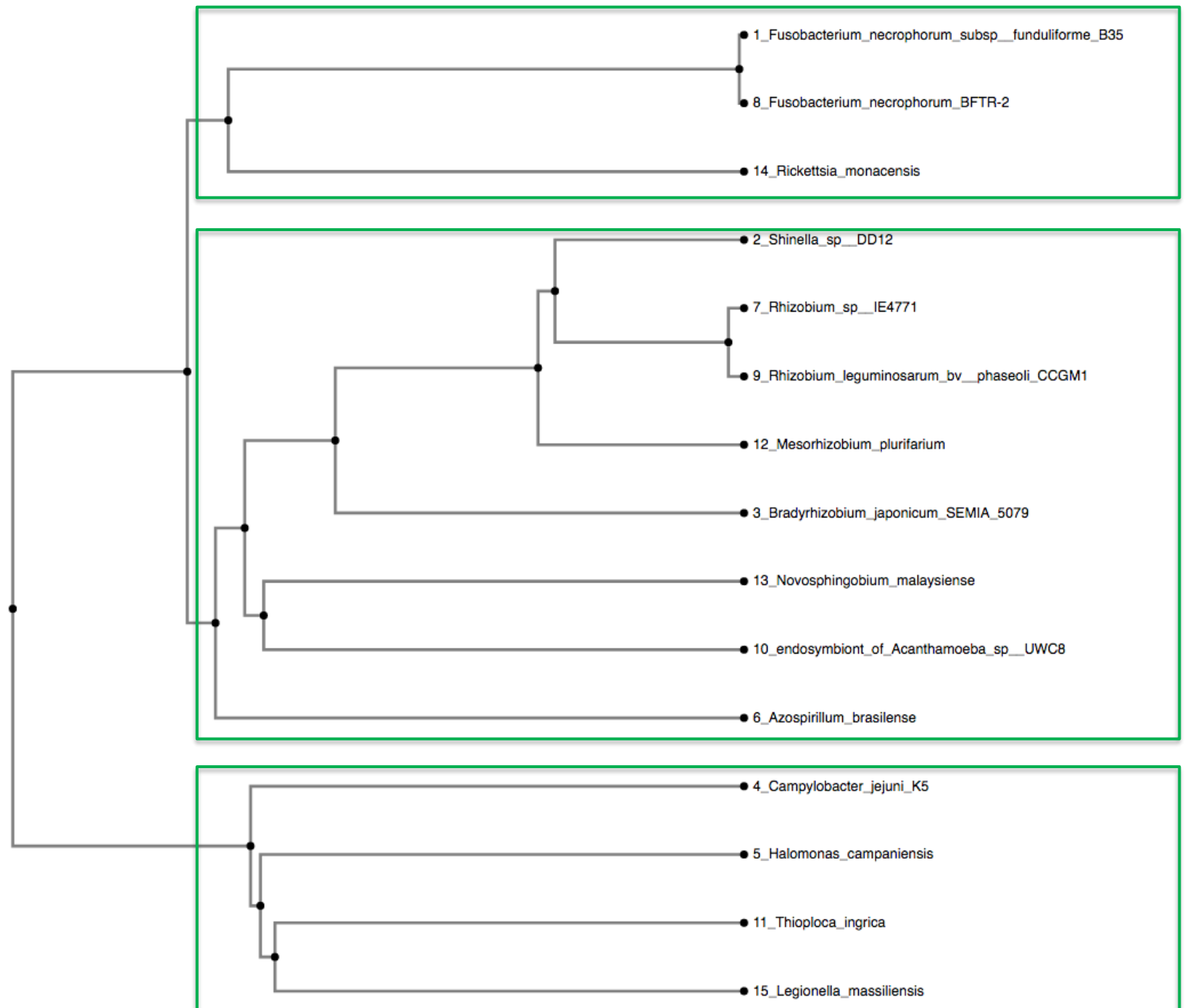
Exercise 2 – Questions

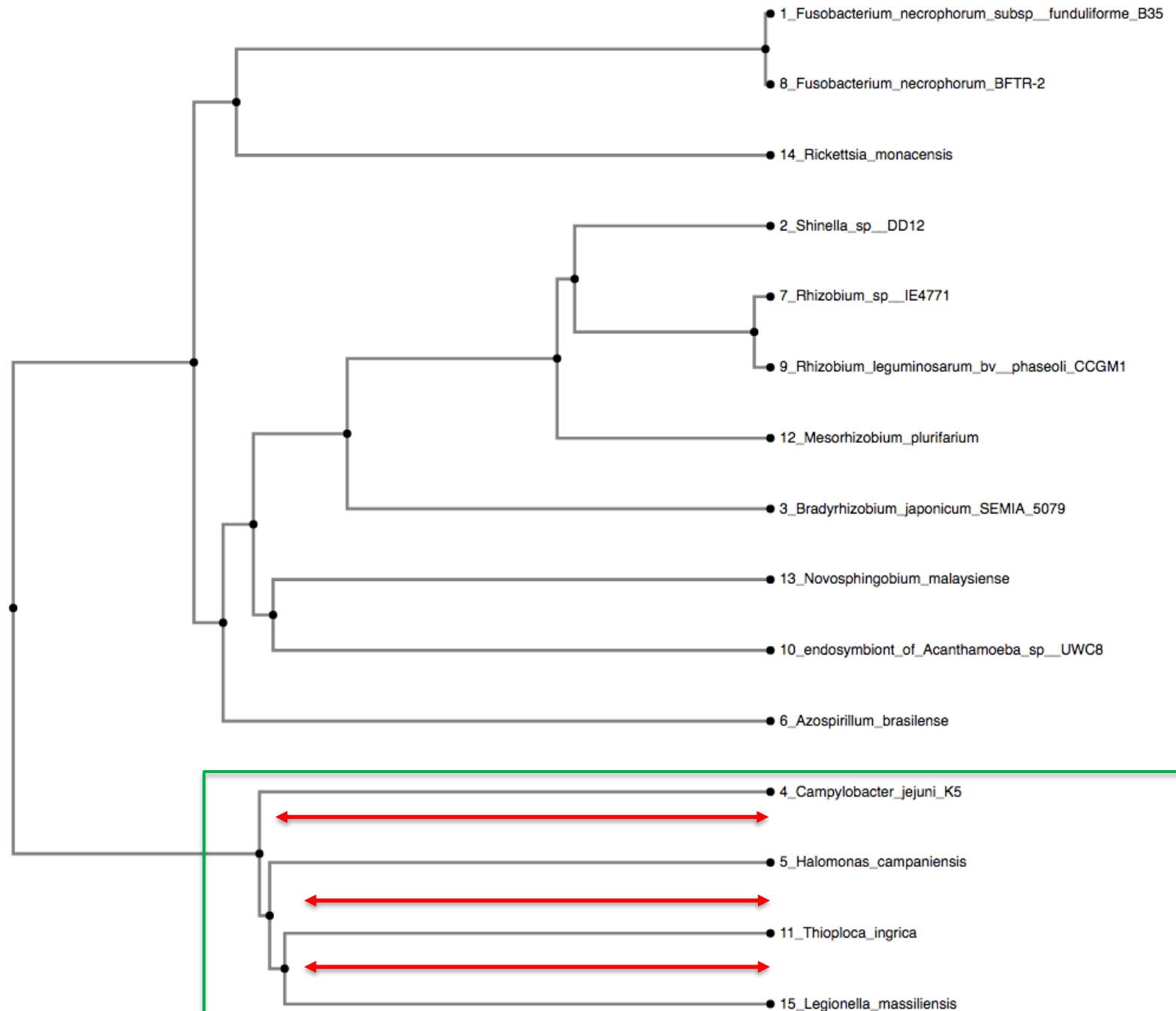
[illegible]



Exercise 3 – Questions







Fusobacterium necrophorum subsp. funduliforme B35

Fusobacterium

Genus

necrophorum

Species

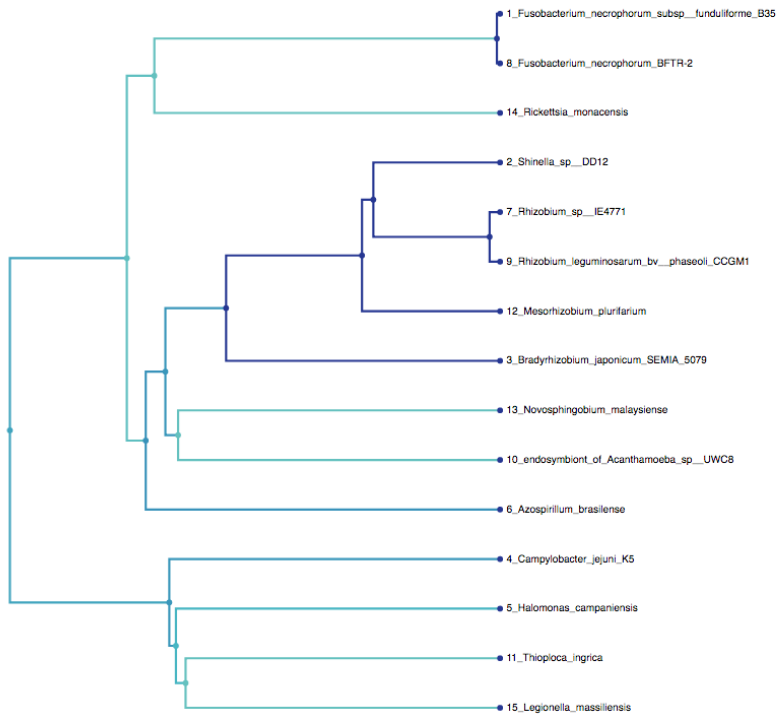
subsp. funduliforme

Subspecies

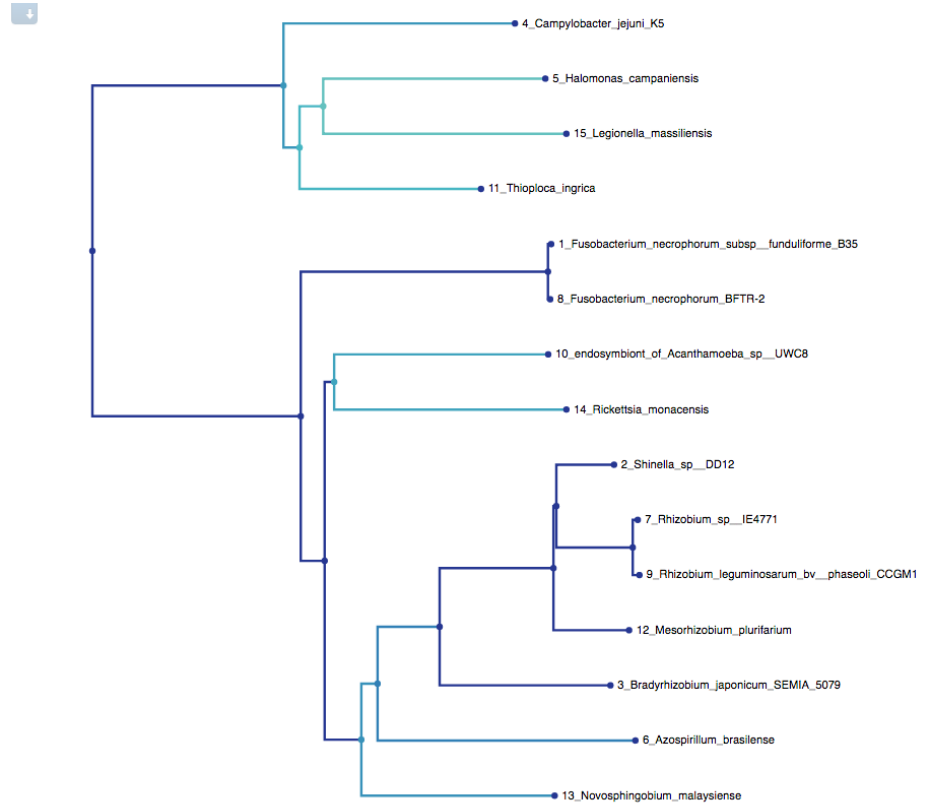
B35

Type strain

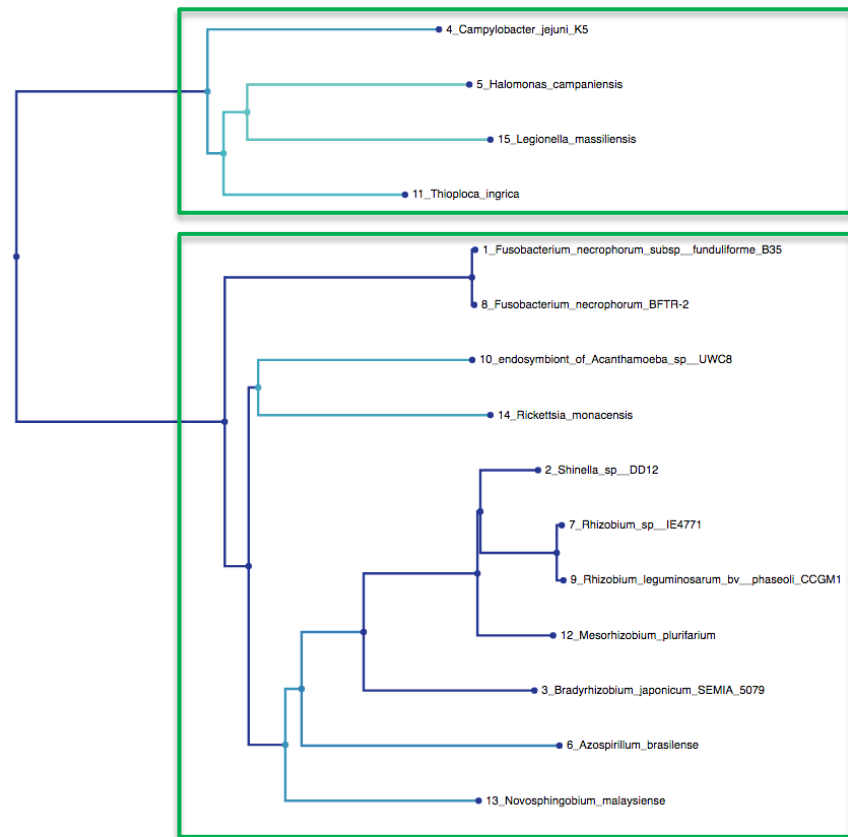
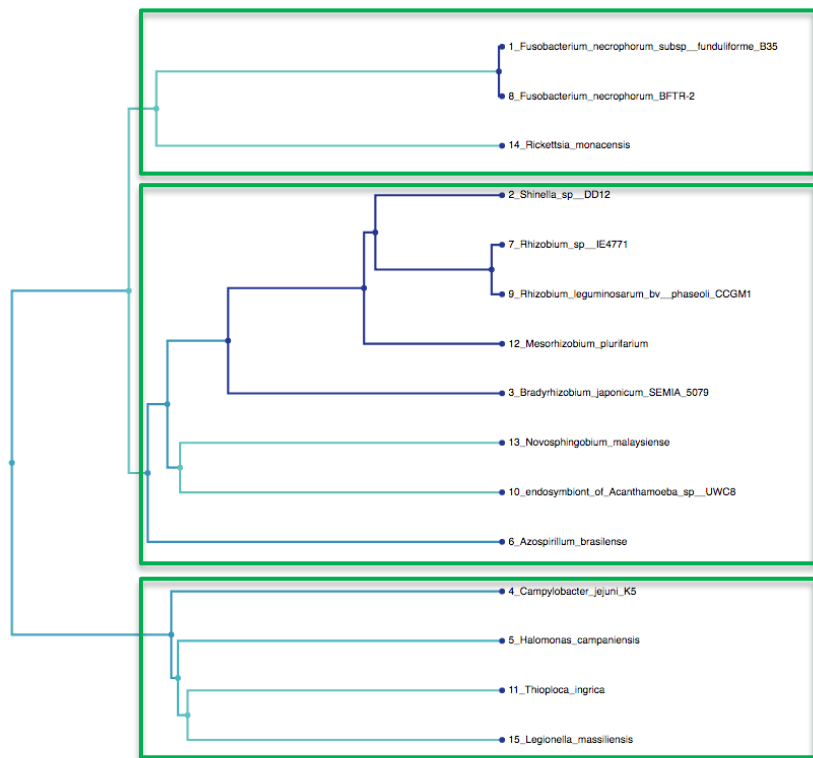
Exercise 4 – Questions

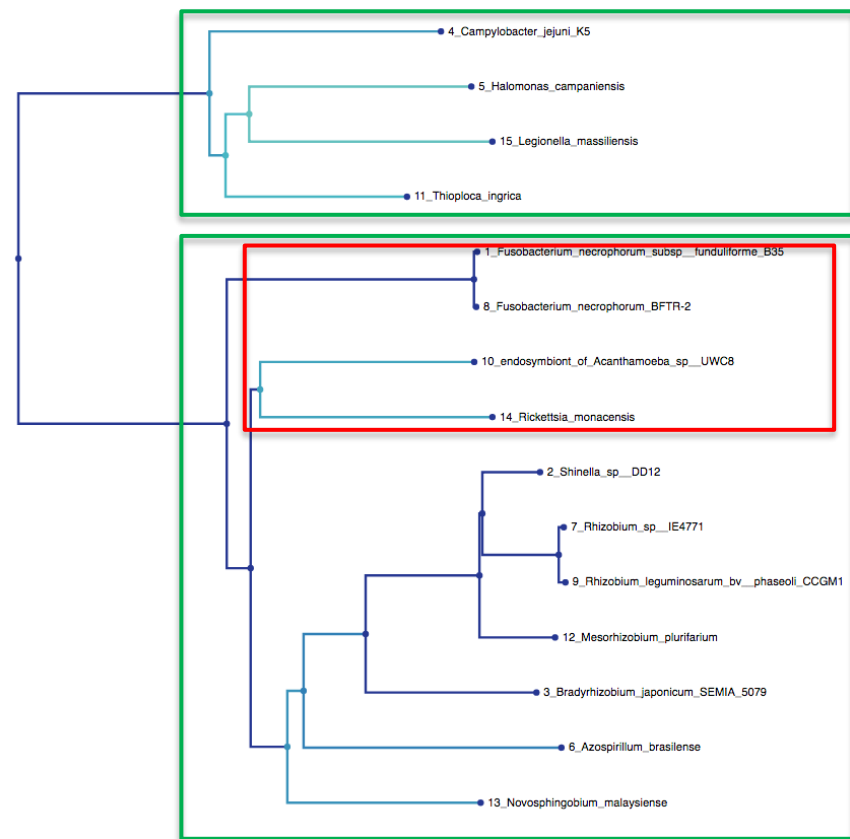
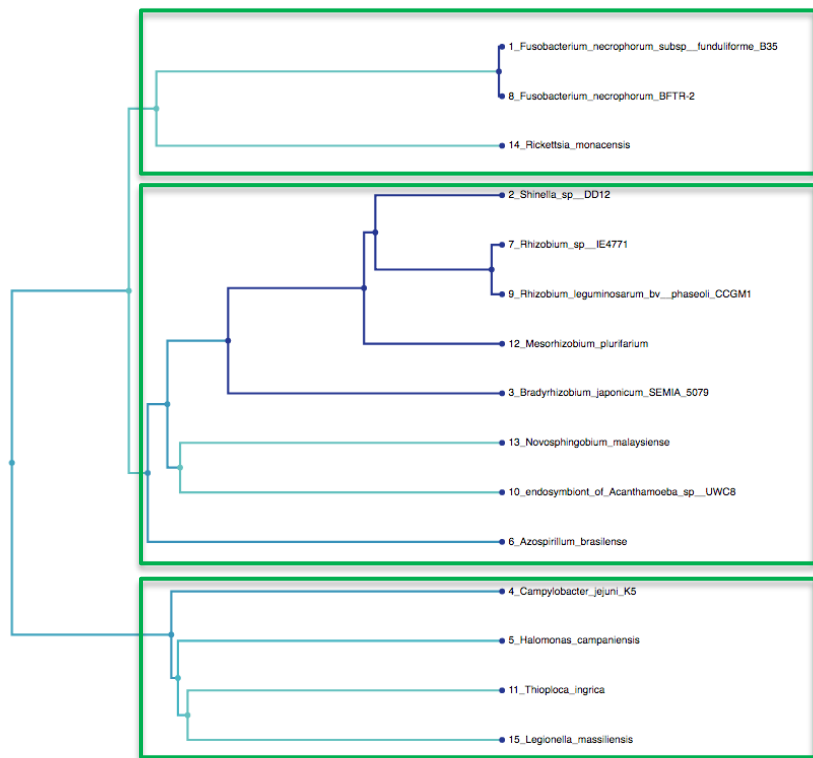


UPGMA



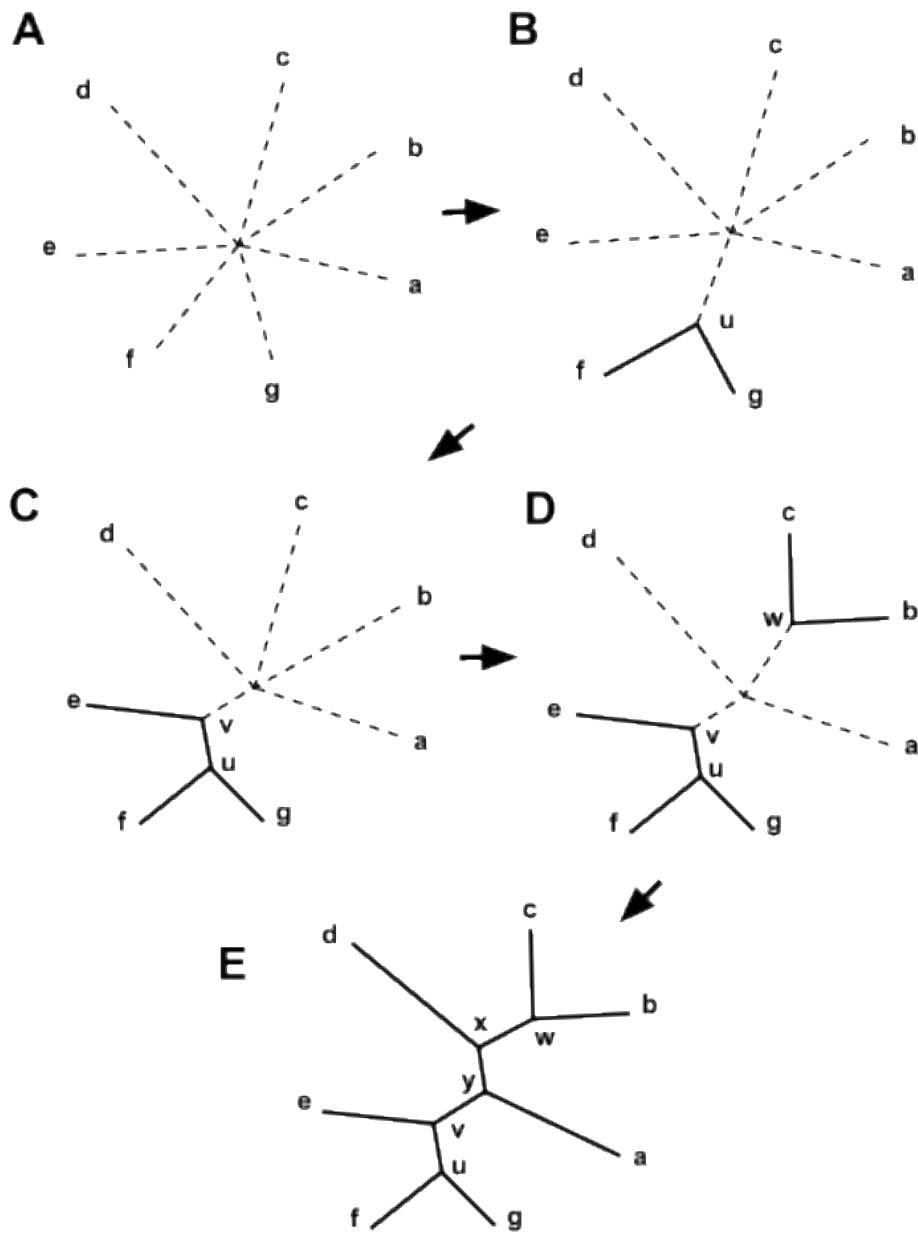
NJ





Exercise 4 – Questions

- uniform molecular clock assumption by UPGMA
- NJ takes distances between 2 sequences and all others



$$Q(i, j) = (n - 2)d(i, j) - \sum_{k=1}^n d(i, k) - \sum_{k=1}^n d(j, k)$$